⇒ Mth632 (Practice Question)

Question NO:01

Find the sum of two complex numbers, $z = 2+i$, $w = -3+9i$

$$z+w = 2+i-3+9i$$
$$= 10i - 1$$

Question NO:02.

Diff $z_1 = -5+i$, $z_2 = -3+9i$

$$z_1 - z_2 = -5+i+3-9i$$
$$= -2-8i$$

Question NO:03

Product: $z_1 = -3+2i$, $z_2 = -1+11i$

$$z_1 z_2 = (-3+2i)(-1+11i)$$
$$= +3-33i-2i+22i^2$$
$$= 3-22-35i$$
$$= -19-35i$$

Question No:04

Divide:- $z_1 = -8+i$, $z_2 = -3+8i$

$$\frac{z_1}{z_2} = \frac{-8+i}{-3+8i} \times \frac{-3-8i}{-3-8i}$$
$$= \frac{(-8+i)(-3-8i)}{9+64}$$

$$\frac{z_1}{z_2} = \frac{24 + 64i - 3i - 8i^2}{73}$$

$$= \frac{24 + 8 + 61i}{73}$$

$$= \frac{32 + 61i}{73}$$

Question NO:05

Simplify:

$$(\sqrt{3} + i3) - i(5 - \sqrt{3}i)$$

$$= \sqrt{3} + i3 - 5i + \sqrt{3}i^2$$

$$= \sqrt{3} + 3i - 5i - \sqrt{3}$$

$$= -2i$$

Question NO:06

Re(z)=? Im(z)=?

$$z = -3 + 9i$$

$Re(z) = -3$, $Im(z) = 9$.

Question NO:07

$z_1 = 2 + i$, $z_2 = -3 + 9i$

Commutative law for multiplication

$$z_1 z_2 = z_2 z_1$$

L.H.S → $z_1 z_2 = (2+i)(-3+9i)$

$= -6 + 18i - 3i + 9i^2$

$= -6 + 15i - 9$

$= \boxed{-15 + 15i}$

R.H.S → $z_2 z_1$

$(-3 + 9i)(2 + i) = z_2 z_1$

$\boxed{z_2 z_1 = -15 + 15i}$

L.H.S = R.H.S.

Q#8 $z^{-1} = ?$

$z = -5 + i$

$z^{-} = \frac{1}{-5+i} \times \frac{-5-i}{-5-i}$

$= \frac{-5-i}{25-i^2} \Rightarrow \frac{-5-i}{25+1} \Rightarrow \frac{-5-i}{26}$ Ans--

Q#9 Write down distributive law of multiplication over addition.

$z_1(z_2 + z_3) = z_1 z_2 + z_1 z_3$

Q#10 $|z| = ?$ $z = -7 + 5i$

$|z| = \sqrt{49 + 25} \Rightarrow$

$|z| = \sqrt{74}$ Ans.

Date

Mon Tue Wed Thu Fri Sat

Q#11 $z_1 = 2+i$, $z_2 = -3+9i$

$$= \sqrt{(-3-2)^2 + (9-1)^2}$$

$$= \sqrt{25+64} \Rightarrow \sqrt{89}$$

$= \sqrt{(-3-2)^2 + (9-1)^2}$

$= \sqrt{25+64} \Rightarrow \sqrt{89}$

Practice Question #03. (File)

Q#1 Verify that $|z|^2 = z\bar{z}$

$z = 3+4i$

L.H.S $= |z|^2$

$|z| = \sqrt{3^2+4^2} = \sqrt{9+16} = \sqrt{25} = 5$

$|z|^2 = (5)^2 = 25$

R.H.S $z\bar{z}$

$\bar{z} = 3-4i$

$z\bar{z} = (3+4i)(3-4i)$

$= 9 - 12i + 12i - 16i^2$

$= 9 - 16(-1)$ $\quad \because i^2 = -1$

$= 9 + 16$

$z\bar{z} = 25$

L.H.S = R.H.S.

Q#2. $\overline{z_1 + z_2} = \bar{z}_1 + \bar{z}_2$

$z_1 = 2+i$, $z_2 = 3+9i$

L.H.S = $\overline{z_1 + z_2}$

$z_1 + z_2 = 2+i+3+9i$

$= 5+10i$

$\overline{z_1 + z_2} = 5-10i$

R.H.S $\Rightarrow$ $\bar{z}_1 + \bar{z}_2$

$\bar{z}_1 = 2-i$, $\bar{z}_2 = 3-9i$

$\bar{z}_1 + \bar{z}_2 = 2-i+3-9i$

$\bar{z}_1 + \bar{z}_2 = 5-10i$

L.H.S = R.H.S

Q#3 Show that:

$Re(z) = \frac{z+\bar{z}}{2}$, $Im(z) = \frac{z-\bar{z}}{2i}$

$z = x+iy$ -(1) , $\bar{z} = x-iy$ -(2)

Now add (1) and (2) and we get Re(z)

$z = x+iy$

$\bar{z} = x-iy$

$z+\bar{z} = 2x$ -(A)

and Sbtract (1) and (2)

$z = x+iy$

$-\bar{z} = -x+iy$

$z-\bar{z} = +2iy$ -(B)

From (A) and (B)

(A) → $\bar{z} + z = 2x$, (B) → $z - \bar{z} = 2iy$

NOW;

$$\boxed{x = \frac{z + \bar{z}}{2}}, \quad \boxed{y = \frac{z - \bar{z}}{2i}}$$

Proved.

Q#4:- $Re(iz) = -Im(z)$

and $Im(iz) = Re(z)$

L.H.S $Re(iz) = -Im(z)$

$Re(i(x+iy)) \Rightarrow Re(ix + i^2 y)$

$= Re(ix - y)$

$\boxed{= -y}$

NOW;

R.H.S $-Im(z) = -Im(x+iy)$

$= -Im - (+y)$

$\boxed{= -y}$

Hence L.H.S = R.H.S

NOW $Im(iz) = Re(z)$.

$Im(iz) = Im(i(x+iy))$

$= Im(ix + i^2 y)$

$= Im(ix - y)$

Im(iz

Now;

Re(

Re(

L.H

Q#5 |z

z, + z,

|z, + z

|z, +

NOW

|z,|

|z,|

|z,|

|z,|

Hen

|z

Pro

$$Im(iz) = x$$

Now;

$$Re(z) = Re(a+iy)$$

$$Re(z) = a$$

L.H.S = R.H.S

Q#5 $|z_1+z_2| \leq |z_1| + |z_2|$

$z_1 = -8+i$, $z_2 = -3+8i$

$$z_1+z_2 = -8+i-3+8i$$

$$= -11+9i$$

$$|z_1+z_2| = \sqrt{(-11)^2+9^2} = \sqrt{121+81}$$

$$|z_1+z_2| = \sqrt{202} = 14.212$$

Now; $|z_1|+|z_2|$

$$|z_1| = \sqrt{8^2+1^2} \Rightarrow \sqrt{64+1} = \sqrt{67}$$

$$|z_2| = \sqrt{9+64} = \sqrt{73}$$

$$|z_1|+|z_2| = \sqrt{67}+\sqrt{73}$$

$$|z_1|+|z_2| = 16.729$$

Hence

$14.212 \leq 16.729$

Proved

Q#6 $z = -7+5i$ in polar coordinates:

$r = |z| = \sqrt{(-7)^2+(5)^2}$

$= \sqrt{49+25}$

$r = \sqrt{74}$

$\tan\theta = \frac{y}{x} \Rightarrow \theta = \tan^{-1}\left(\frac{y}{x}\right)$

$\theta = \tan^{-1}\left(\frac{5}{-7}\right)$

~~$\theta = \tan$~~ $\boxed{\theta = -35.5}$

in polar coordinates;

$z = r(\cos\theta + i\sin\theta)$

$\boxed{z = \sqrt{74}\left(\cos(-35.5)+i\sin(-35.5)\right)}$

Q#7: $z = (0, 11) = 11i$ in polar rep.

when $x = 0$ then following rule:- $\theta = \frac{\pi}{2}$ if $Im(z) > 0$

Here $Im(z) = 11 > 0$

so; $\theta = \pi/2$.

$r = \sqrt{0^2 + 121^2} = \sqrt{121}$

$r = 11$

in polar form,

$z = 11\left[\cos\frac{\pi}{2} + i\sin\frac{\pi}{2}\right]$

$z = 11\left[0 + i\right]$

$z = 11i$